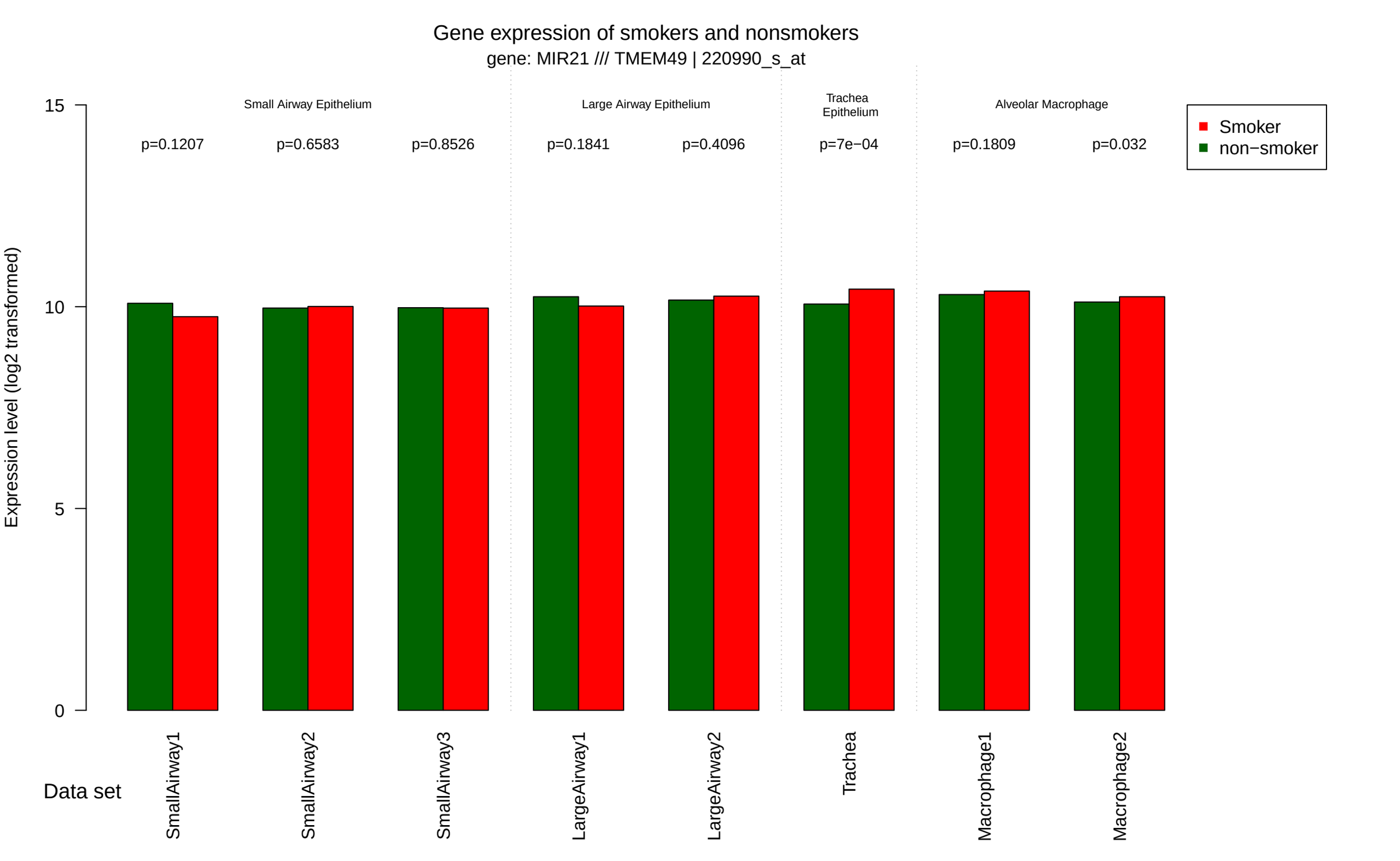
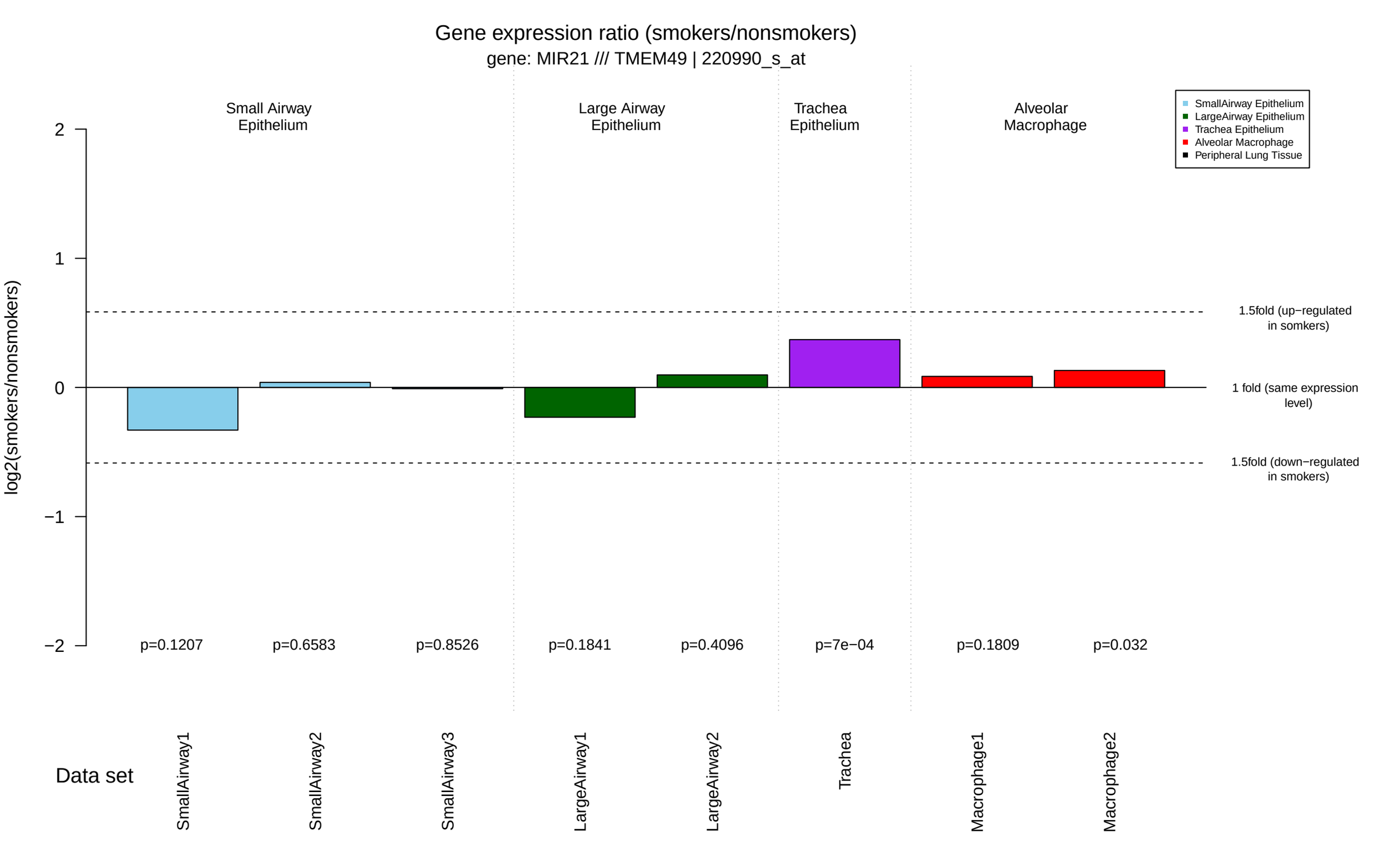
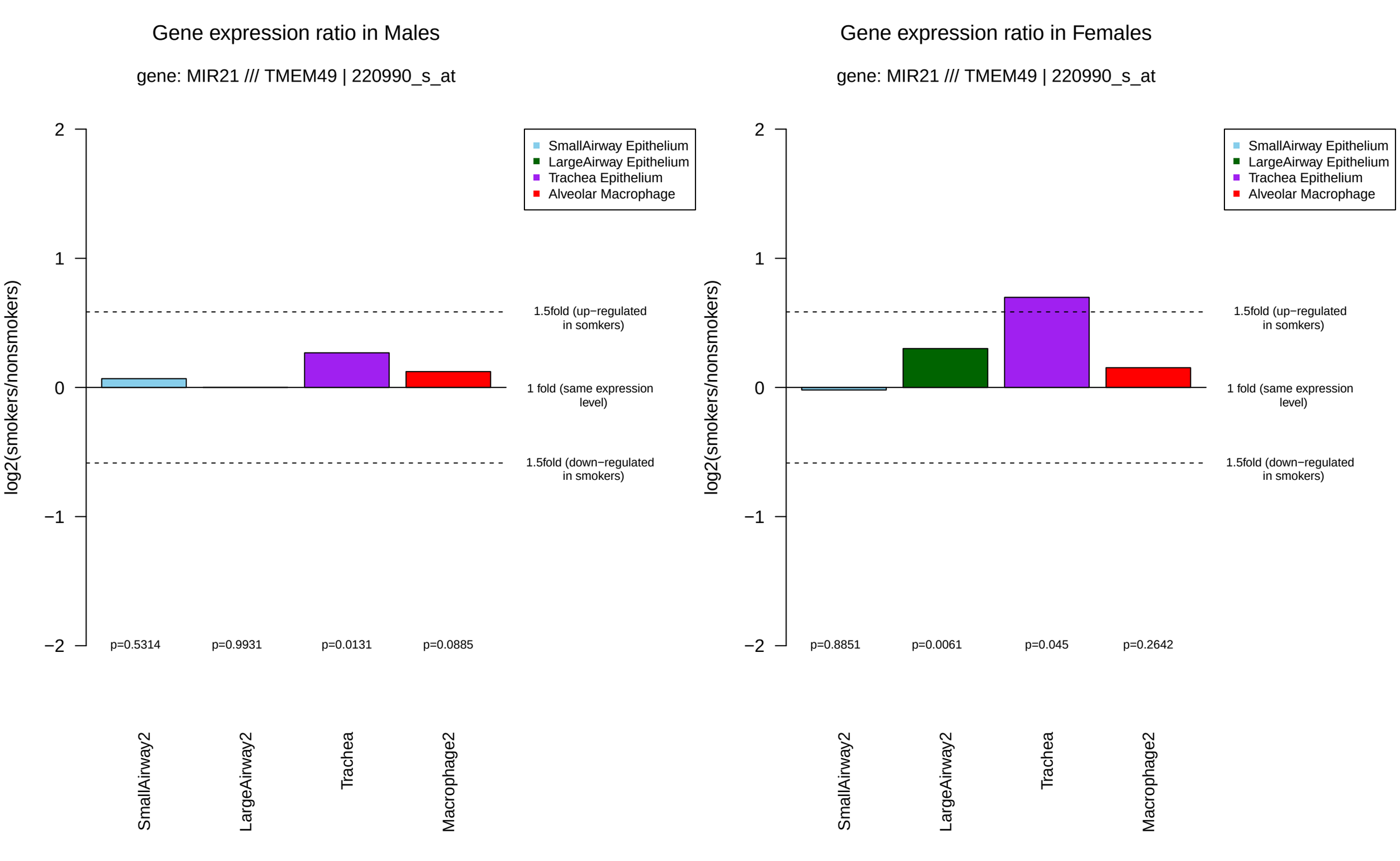
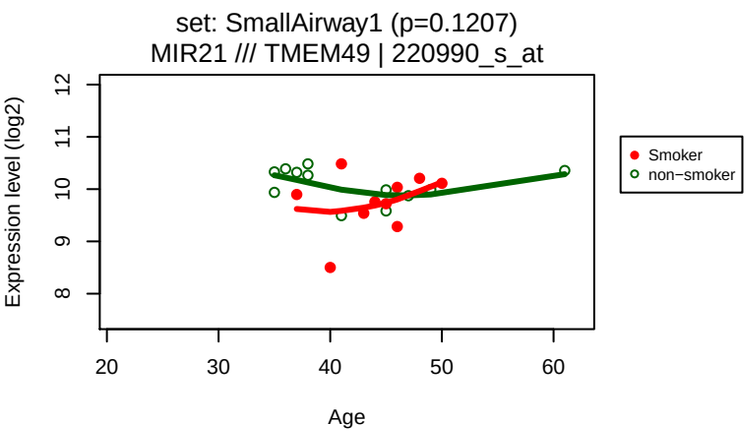
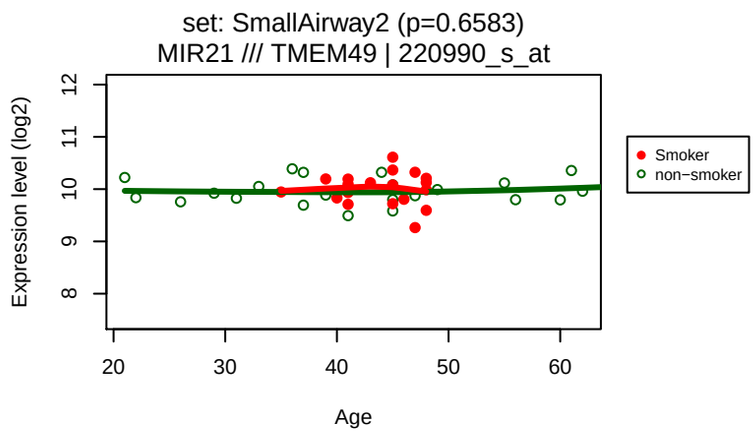
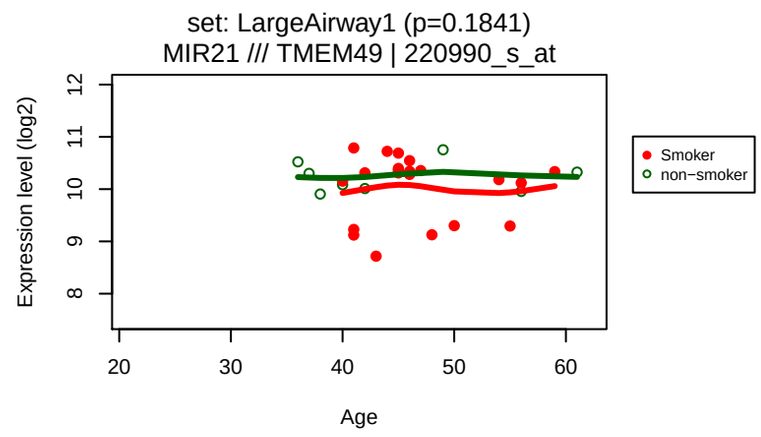

Gene expression of smokers and nonsmokers
gene: MIR21 /// TMEM49 | 220990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1207
p=0.6583
p=0.8526
p=0.1841
p=0.4096
p=7e−04
p=0.1809
p=0.032
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

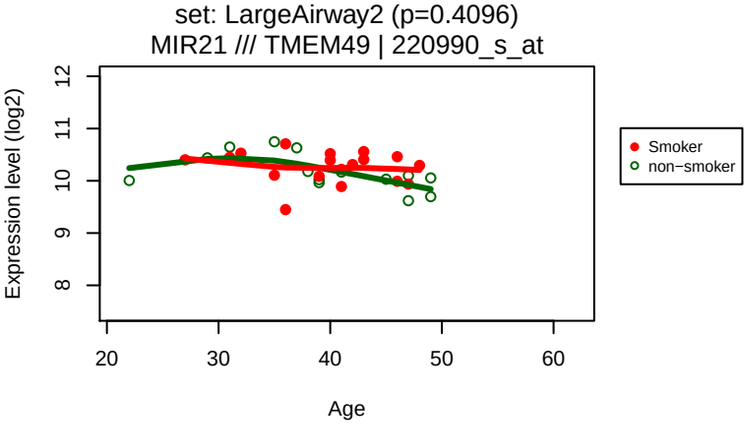

Gene expression ratio (smokers/nonsmokers)
gene: MIR21 /// TMEM49 | 220990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1207
p=0.6583
p=0.8526
p=0.1841
p=0.4096
p=7e−04
p=0.1809
p=0.032
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

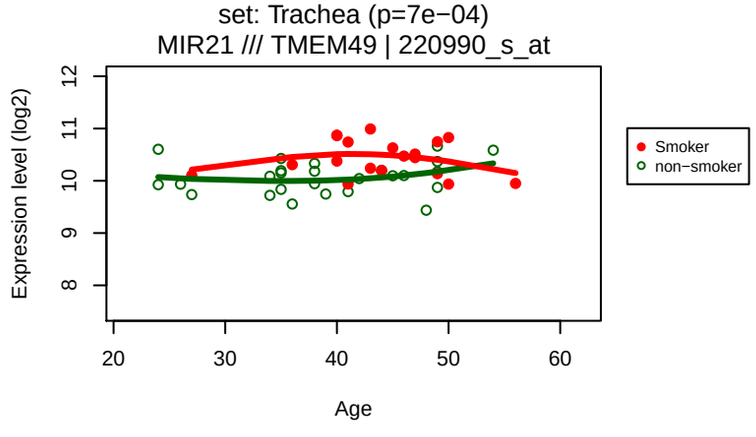

Gene expression ratio in Males
gene: MIR21 /// TMEM49 | 220990_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5314
p=0.9931
p=0.0131
p=0.0885
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MIR21 /// TMEM49 | 220990_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8851
p=0.0061
p=0.045
p=0.2642
SmallAirway2
LargeAirway2
Trachea
Macrophage2

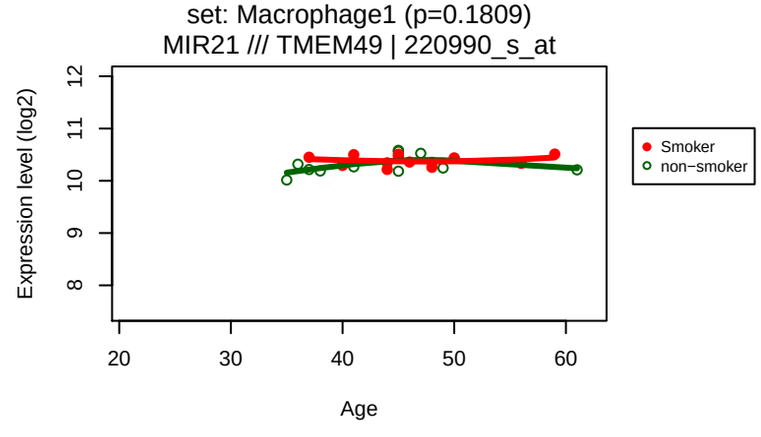

set: SmallAirway1 (p=0.1207)
MIR21 /// TMEM49 | 220990_s_at
Expression level (log2)
Age
Smoker
non−smoker

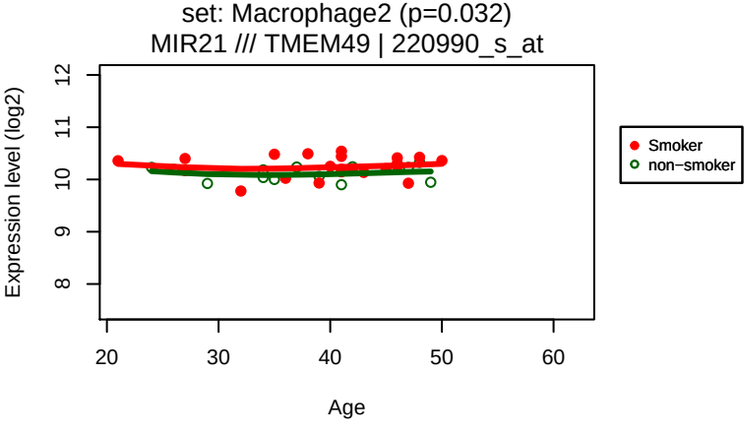

set: SmallAirway2 (p=0.6583)
MIR21 /// TMEM49 | 220990_s_at
Expression level (log2)
Age
Smoker
non−smoker

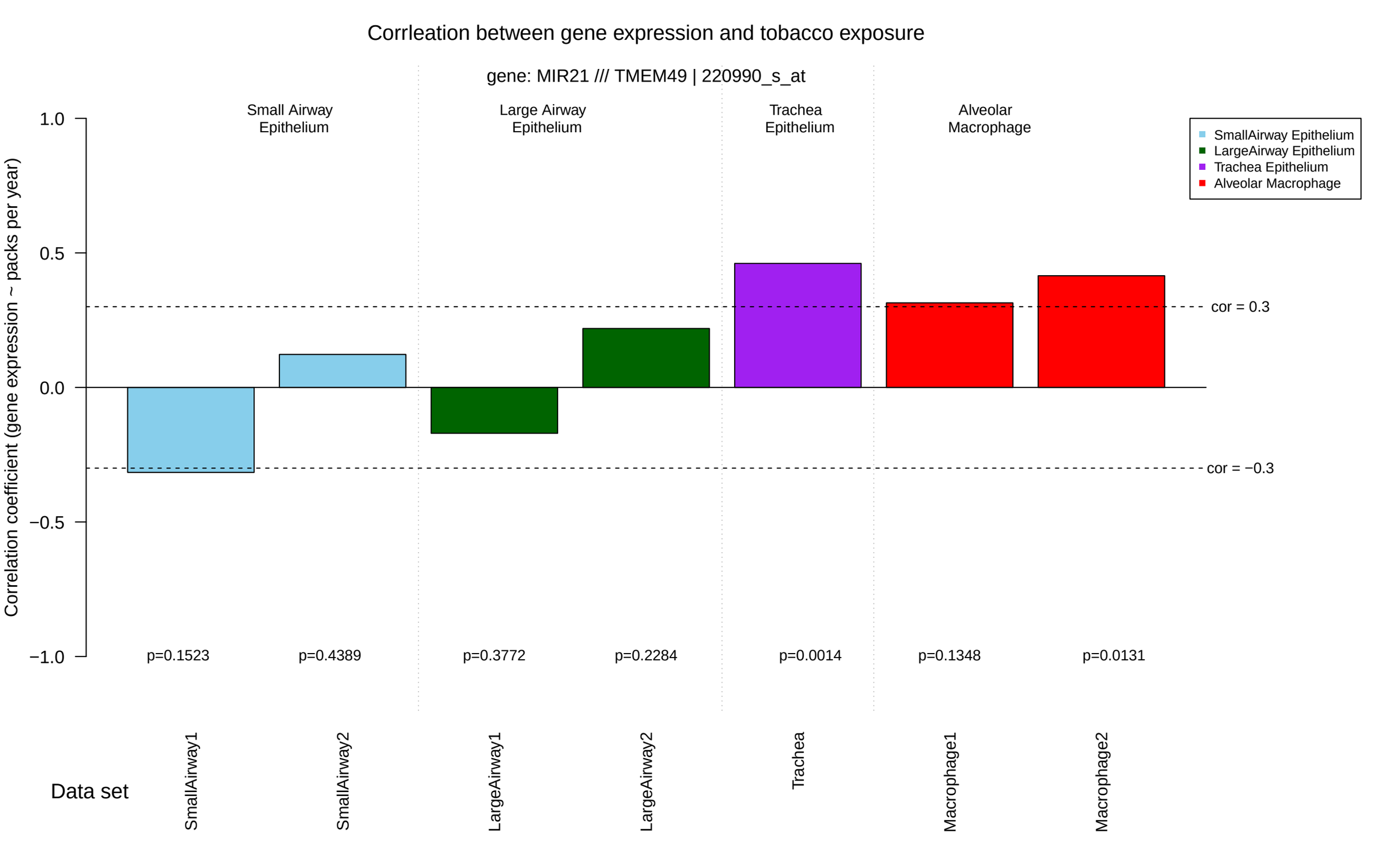

set: LargeAirway1 (p=0.1841)
MIR21 /// TMEM49 | 220990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4096)
MIR21 /// TMEM49 | 220990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
MIR21 /// TMEM49 | 220990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1809)
MIR21 /// TMEM49 | 220990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.032)
MIR21 /// TMEM49 | 220990_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MIR21 /// TMEM49 | 220990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1523
p=0.4389
p=0.3772
p=0.2284
p=0.0014
p=0.1348
p=0.0131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2